

Rectal laterally spreading tumors successfully treated in two steps by endoscopic submucosal dissection and endoscopic mucosal resection

Background Endoscopic submucosal dissection (ESD) is an advanced technique of therapeutic endoscopy alternative to endoscopic mucosal resection (EMR) for superficial gastrointestinal neoplasms >2 cm. ESD allows for the direct dissection of the submucosa and large lesions can be resected en bloc. ESD is not limited by resection size, increases histologically complete resection rates and may reduce the local recurrence. Nevertheless, the technique is time-consuming, technically demanding and associated with a high complication rate. To reduce the risk of complications, different devices and technical advances have been proposed with conflicting results and, still, ESD en bloc resections of huge lesions are associated with increased complications. **Case Presentation** We successfully used a combined ESD/EMR technique for huge rectal laterally spreading tumors (LSTs). ESD was used for circumferential resection of 2/3 of the lesion followed by piecemeal resection (2-3 pieces) of the central part of the tumour. In all three patients we obtained the complete dissection of the polyp and the complete histological evaluation in absence of complications and recurrence at 6 months' follow up. **Conclusions** In the treatment of rectal LSTs, the combined treatment - ESD/EMR resection may be considered a suitable therapeutic option, indicated in selected cases as an alternative to surgery, in which the two techniques are neither reliable nor safe separately. However, to confirm our results, larger trials with longer follow up are required together with improvement of the technique and of the technical devices.